ALGEBRA 1 · STATISTICS AND DATA MODELING

Line of Best Fit Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Line of Best Fit Lab** on Mr. K's Classroom: Algebra 1 → Statistics and Data Modeling → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Hours studied and test scores: positive, negative or no correlation?

   Positive: more hours tends to mean higher scores.

2. What do you think happens to r if a student studied 8 hours but scored 40?

   r gets smaller (the correlation gets weaker).

2 Explore use the tool and record what you see

Correlation

Record r from the tool.

Situation	r	Strength and direction
Original data	**0.967**	**strong positive**
After adding the outlier	**0.370**	**weak positive**
Points dragged to make r negative	**answers vary (below 0)**	**negative**
Points dragged to make r near 0	**answers vary (−0.2 to 0.2)**	**no clear correlation**

Your line vs the best line (original data)

Drag your line, then show the line of best fit.

	Equation	Total squared error
Your line	**answers vary**	**answers vary**
Line of best fit	**y = 4.67x + 49.3**	**about 72**

3 Write the rule fill in each blank from what you observed

1. The correlation coefficient r is always between **−1** and **1**.
2. r close to 1 means a strong **positive** correlation; close to −1 a strong **negative** correlation; close to 0 **little or no correlation**.
3. A residual is the **vertical** distance from a point to the line: actual − predicted.
4. The line of best fit makes the sum of the squared residuals as **small** as possible.
5. An outlier can **weaken (lower)** the value of r and pull the line toward itself.

4 Check no tool: show your work

1. Use y = 4.67x + 49.3 to predict the score for 9 hours of study.

   about 91

2. What does the slope 4.67 mean in this situation?

   Each extra hour of study adds about 4.67 points

3. A student studied 5 hours and scored 68. Find the residual.

   68 – 72.65 ≈ –4.65

4. r = 0.9 for ice cream sales and sunburns. Does ice cream cause sunburns?

   No. Correlation is not causation; hot weather drives both.

5 Reflect

Why do we square the residuals instead of just adding them?

Positive and negative residuals would cancel; squaring makes every miss count.